

510(k) SUMMARY
Cedars-Sinai Medical Center's Autoplaque 3.0

In accordance with 21 CFR § 807.92 the following summary of information is provided:

I. Submitter

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II. Date of 510(k) Summary Preparation:

May 2, 2023

III. Device

Device Proprietary Name:	Autoplaque 3.0
Common or Usual Name:	Image Processing System, Radiological
Classification Name:	Automated Radiological Image Processing Software Radiological Imaging Processing System
Regulation Number:	21 CFR § 892.2050
Product Code:	QIH, LLZ
Device Classification	II

IV. Predicate Device

Primary Predicate Device:

- Autoplaque add-on ORS Visual, K122429, Object Research Systems (ORS) Inc.

V. Device Description

Autoplaque 3.0, a stand-alone software, performs post-processing of coronary Computed Tomography Angiography (CTA) images and measurements of components of images using computerized algorithms.

Autoplaque 3.0 aids the physician with measurement of coronary artery stenosis and provides measurements for coronary plaque and coronary artery remodeling. Autoplaque 3.0 does not replace standard clinical practice or clinician decision making.

Autoplaque 3.0 allows for standardized characterization of plaque and stenosis from DICOM image data (loaded from the local computer hard drive) and includes the following features:

- Review of heart and coronary vessels in Multiplanar Reformatting (MPR), curved MPR, and straightened MPR views;
- Measurement of vessel diameter and area;
- Characterization and measurement of plaque parameters; and
- Measurement of lumen diameter, area, and luminal stenosis.

Autoplaque 3.0 includes automated vessel, plaque and lumen segmentation, which is reviewed and can be edited, if necessary, by the clinician.

Autoplaque 3.0 can run on Windows or Mac computer platforms. The minimum hardware specifications are specified in the user manual.

VI. Indications for Use

Autoplaque is intended to provide an optimized non-invasive application to analyze coronary anatomy and pathology and aid in determining treatment paths from a set of Computed Tomography (CT) Angiographic images.

Autoplaque is a workstation-based post processing application. It is a non-invasive diagnostic reading software intended for use by cardiologists and radiologists as an interactive tool for viewing and analyzing cardiac CT data for determining the presence and extent of coronary plaques and luminal stenoses.

The software is not intended to replace the skill and judgment of a qualified medical practitioner and should only be used by people who have been appropriately trained in the software's functions, capabilities and limitations. Users should be aware that certain views make use of interpolated data. This is data that is created by the software based on the original data set. Interpolated data may give the appearance of healthy tissue in situations where pathology that is near or smaller than the scanning resolution may be present.

Autoplaque must be installed on a suitable commercial computer platform. It is the user's responsibility to ensure the monitor quality and ambient light conditions are consistent with the clinical applications.

Typical users of Autoplaque are trained medical professionals, including but not limited to radiologists, clinicians, technologists, and others.

VII. Comparison of Technological Characteristics

Autoplaque 3.0 and the predicate device are identical with respect to intended use and technological characteristics with the exception of the following:

- Compatibility of the subject device with Mac operating system (OS);
- Provision of the subject device as stand-alone software versus an add-on; and
- Deep learning-based vessel, plaque and lumen segmentation, which is reviewed and can be edited, if necessary, by the clinician.

The table below compares the key technological features between the subject and predicate devices.

Parameter	Subject Device: Autoplaque 3.0 (K212758)	Predicate Device: Autoplaque add-on ORS Visual (K122429)
Computer operating system	Windows OS Mac OS	Windows OS only
Stand-alone software	Yes	No, ORS Visual add-on
DICOM compliance	DICOM 3.1	Same
2D Imaging	Review of coronary vessels in 2D MPR, curved MPR, and straightened view.	Same
2D Measurement	2D measurement tools of vessel diameter and contour	Same
3D Imaging	Review of structures in 3D	Same
Maximum intensity projection (MIP)	MIP with interactive control	Same
Multiplanar reformatting (MPR)	MPR with oblique slicing and variable slab thickness	Same
<i>Quantitative Measurements</i>		
NCP volume: non-calcified plaque volume	Yes	Same
CP volume: calcified plaque volume	Yes	Same
LD-NCP volume: low-density noncalcified plaque volume	Yes	Same
Total plaque volume	Yes	Same
Vessel volume	Yes	Same

Parameter	Subject Device: Autoplaque 3.0 (K212758)	Predicate Device: Autoplaque add-on ORS Visual (K122429)
NCP burden: noncalcified plaque volume/analyzed vessel volume	Yes	Same
LD-NCP burden: low-density noncalcified plaque volume/analyzed vessel volume	Yes	Same
CP burden: calcified plaque volume/analyzed vessel volume	Yes	Same
Total plaque burden: total plaque volume/analyzed vessel volume	Yes	Same
Plaque composition NCP: Noncalcified plaque composition (NCP volume /total plaque volume)	Yes	Same
Plaque composition CP: Calcified plaque composition (CP volume/total plaque volume)	Yes	Same
Plaque composition LD-NCP: low-density noncalcified plaque composition (LD-NCP volume/ NCP volume)	Yes	Same
Diameter stenosis: maximal diameter stenosis, with respect to proximal and distal references	Yes	Same
QCAD: Maximal diameter stenosis, with respect to proximal and distal references	Yes	Same
Remodeling index: ratio of maximum vessel area/proximal and distal references	Yes	Same
Area stenosis: maximum area stenosis, with respect to proximal and distal references	Yes	Same
Plaque length: diseased vessel length	Yes	Same
Contrast density difference: maximum difference in contrast density over lesion with respect to proximal	Yes	Same
MLD: minimal luminal diameter over lesion	Yes	Same

Parameter	Subject Device: Autoplaque 3.0 (K212758)	Predicate Device: Autoplaque add-on ORS Visual (K122429)
MLA: minimum luminal area over lesion	Yes	Same
Vessel profile: Area, maximum diameter, minimum diameter measured from selected vessel cross section	Yes	Same
Lumen profile: Area, maximum diameter, minimum diameter measured from selected lumen cross section	Yes	Same
Vessel, plaque, and lumen segmentation	Deep learning based, validated with ground truth by clinician expert readers, with full option to edit.	Algorithm based, validated with ground truth by clinician expert readers, with full option to edit.

Discussion: As shown above, there are slight technological differences between the subject and predicate devices with respect to computer operating system, provision of the product as stand-alone software, and vessel, plaque, and lumen segmentation method; however, these differences do not raise new questions of safety and effectiveness, and the differences are addressed through software verification and validation as well as clinical performance studies.

VIII. Performance Data

Software verification and validation were conducted on the subject device to demonstrate that the subject device meets specifications and works as intended.

In addition, clinical validation studies were undertaken to verify and validate that Autoplaque 3.0 performs as intended. A summary of these studies is provided below.

1. Comparison with Expert Reader Measurements in a Multicenter US Patient Population

Performance data was collected on a data set of clinically indicated coronary computed tomography angiography (CCTA) images from multiple U.S. sites. The patient population consisted of 201 patients (53% men, 47% women) with a mean age of 60 ± 12 years. A total of 781 lesions from 201 patients were analyzed. Ethnicity available in 80 patients was as follows: White alone (not Hispanic or Latino) 18%, Black 19%, Hispanic or Latino 61%, and American Indian and Alaska native alone 1%. The CT scanners to collect images included commercial CT scanners from Siemens, Philips and GE. Ground truthing was performed by a board-certified imaging cardiologist. This study established concordance and agreement between clinician expert reader and Autoplaque 3.0 measurements for total plaque volume, calcified plaque volume, non-calcified plaque volume, and diameter stenosis using the acceptance criteria of an intraclass correlation coefficient and correlation coefficient. An excellent agreement was

shown between the subject device and expert reader measurements for total plaque volume, noncalcified plaque volume, calcified plaque volume, and diameter stenosis.

2. Comparison with Expert Reader Measurements in a Non-US Patient Population

Performance data was collected on clinically indicated coronary computed tomography angiography (CCTA) images from multiple non-U.S. external sites. The patient population consisted of 175 patients (53% men, 47% women) with a mean age of age 56 ± 9 years. A total of 1081 lesions from 175 patients were analyzed. The CT scanners to collect images included commercial CT scanners from Siemens, Philips and Canon. Ground truthing was performed by two cardiologists with one additional highly experienced radiologist to resolve discrepancies. This study established concordance and agreement between clinician expert reader and Autoplaque 3.0 measurements for total plaque volume, calcified plaque volume, non-calcified plaque volume, and diameter stenosis using the acceptance criteria of an intraclass correlation coefficient and correlation coefficient. An excellent agreement was shown between the subject device and expert reader measurements for total plaque volume, noncalcified plaque volume, calcified plaque volume, and diameter stenosis. In addition, the average per-patient plaque analysis time was 5.7 seconds for the subject device versus 25-30 minutes taken by experts.

3. Comparative Study with the Predicate Device

To establish that the subject device provides clinical performance for total plaque volume, calcified plaque volume, non-calcified plaque volume, and diameter stenosis that is substantially equivalent to the predicate device, a total of 30 lesions (from 27 patients) throughout the coronary artery tree were analyzed using both the subject and predicate devices. The patient population were 60% men and 40% women with a mean of age 64.5 ± 10.1 years. The CT scanner used to collect the images was a commercial Siemens CT scanner. There was an excellent correlation between the subject and predicate devices for total plaque volume, non-calcified volume, calcified plaque volume, and diameter stenosis. These measurements also demonstrated excellent intraclass correlation agreement. It took less than 2 seconds per lesion for the subject device Autoplaque 3.0 to analyze, which was much faster than the predicate device (30 seconds per lesion).

IX. Conclusion

The information provided above supports that Autoplaque 3.0 is as safe and effective as the predicate device. Although there are minor differences between the subject and predicate devices, the differences are addressed through software verification and validation studies, and performance testing. Therefore, it is concluded that Autoplaque 3.0 is substantially equivalent to the predicate device.